

Class 6 Science Olympiad Syllabus (As per CBSE Curriculum)

SECTION 1: Life Science – The Living World

- Characteristics of living organisms
 - Classification of living beings: plants & animals (kingdom intro)
 - Cell – basic unit of life (intro with diagram)
 - Structure and functions of plant and animal cells
 - Photosynthesis and respiration in plants
 - Reproduction in plants – seeds, germination, and dispersal
 - Human body systems:
 - Digestive system
 - Respiratory system
 - Circulatory system
 - Skeletal and muscular systems
 - Health, hygiene, diseases and their prevention
 - Food components – carbohydrates, proteins, fats, vitamins, minerals, roughage, water
 - Balanced diet and malnutrition
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SECTION 2: Environmental Science

- Ecosystem – components, food chains, and food webs
 - Adaptations in plants and animals
 - Pollution – types (air, water, land, noise), causes, and prevention
 - Natural resources – renewable and non-renewable
 - Conservation of water, soil, wildlife, and forests
 - Greenhouse effect and global warming (basic awareness)
 - Waste management – biodegradable and non-biodegradable materials
 - Deforestation and its impact
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SECTION 3: Physical Science (Physics & Chemistry Basics)

Matter and Materials

- States of matter – solid, liquid, gas
- Physical and chemical changes
- Solutions, solubility, and separation methods (filtration, evaporation, etc.)
- Properties of materials – transparency, hardness, solubility, conductivity
- Acids, bases, and salts – identification using indicators (intro)

Force, Motion, and Energy

- Types of forces – gravity, friction, magnetic, muscular
- Types of motion – linear, circular, periodic
- Work and energy – concept and examples
- Simple machines – types and uses (lever, pulley, wedge, screw, etc.)
- Types of energy – mechanical, thermal, sound, light
- Conservation of energy (intro)

Light and Sound

- Light – sources, reflection, shadows, mirrors

- Sound – sources, pitch, vibration, noise pollution
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SECTION 4: Earth & Space Science

- Layers of the Earth – crust, mantle, core
 - Rocks and minerals – types and uses
 - Soil – formation, types, erosion and conservation
 - Water – distribution on Earth, water cycle
 - Weather and climate – difference, elements of weather
 - Solar System – planets, Sun, Moon, satellites, asteroids
 - Rotation and revolution of Earth – day & night, seasons
 - Moon phases and tides
 - Constellations and stars – basic introduction
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SECTION 5: Science in Everyday Life

- Scientific inventions and inventors (age-appropriate)
 - Use of science in home, school, transport, and communication
 - Safety measures during experiments and in daily life
 - Importance of clean water, sanitation, and health
 - Introduction to electricity and circuits – conductors, insulators, components
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SECTION 6: Logical Reasoning & HOTS (Higher Order Thinking Skills)

- Number and pattern series
- Odd one out
- Analogies
- Classification and sorting
- Scientific puzzles and reasoning-based questions
- Cause and effect reasoning in scientific scenarios

